

WEEK 6 LAB ACTIVITIES – NETWORK SNIFFING

Practical Hands-On Security Lab

CCS6324: Ethical Hacking & Penetration Testing

IMPORTANT DISCLAIMER

All lab activities in this course must be performed only in a controlled laboratory environment. Unauthorized access to computer networks is illegal under the Computer Fraud and Abuse Act (CFAA) and similar laws in most countries. These techniques should NEVER be used against real systems without explicit written permission from the system owner.

Learning Outcomes

- Master Wireshark for network traffic analysis and packet inspection
- Understand and execute Man-in-the-Middle (MITM) attacks in a controlled environment
- Perform ARP poisoning attacks using Ettercap
- Analyze captured network traffic and extract sensitive information
- Understand network-level attack vectors and their impact
- Learn defensive techniques to detect and prevent sniffing attacks

PART 1: ENVIRONMENT SETUP

Requirements

- Virtual machine software (VirtualBox is free)
- Linux distribution (Ubuntu or Kali Linux recommended)
- At least 4GB RAM and 20GB disk space
- Administrator/root access to your lab system
- Isolated lab network (NOT connected to the internet or your organization's network)

Setting Up Your Lab Network

For these labs, you will create a virtual network with multiple machines:

- Machine 1 (Attacker): Kali Linux or Ubuntu with security tools
- Machine 2 (Victim/Server): Ubuntu or similar Linux system
- Virtual Network: Bridged or NAT network connecting the machines
- Isolated: NOT connected to the host machine's network or internet

Installing Required Tools

On your attacker machine, install the following tools. Run these commands in the terminal:

```
sudo apt-get update
```

```
sudo apt-get install wireshark tshark tcpdump ettercap-graphical mitmproxy -y
```

For macOS users, use Homebrew:

```
brew install wireshark tcpdump mitmproxy
```

For Windows users, download installers from official websites and add to system PATH.

PART 2: WIRESHARK PACKET CAPTURE AND ANALYSIS

What is Wireshark?

Wireshark is a graphical packet analyzer that captures and displays network packets in real-time. It allows you to see the contents of network packets and understand how network protocols work. This is an essential tool for network troubleshooting and security analysis.

Activity 1: Basic Packet Capture

Objective: Capture HTTP traffic and examine packet structure

Step 1: Launch Wireshark

On Linux:

```
sudo wireshark &
```

On Windows/macOS: Open Wireshark from the Applications menu.

Step 2: Select Network Interface

- Look for the 'Interface List' on the left side of Wireshark
- Select your main network interface (usually eth0, en0, or your WiFi adapter)
- Double-click to start capturing packets

Step 3: Generate Traffic

Open a web browser and visit a website. It will generate HTTP traffic that Wireshark will capture.

Visit: `http://example.com` (not HTTPS to capture readable traffic)

Step 4: Stop Capture

- Click the red square (stop) button in the toolbar
- You should now see packets in the display

Step 5: Analyze the Packets

The Wireshark window has three panes:

- Top Pane: List of all captured packets
- Middle Pane: Details of the selected packet showing all protocol layers
- Bottom Pane: Raw hexadecimal data of the selected packet

Step 6: Apply Filters

Filters help you find specific traffic. In the filter bar at the top, type a filter and press Enter:

```
http      (shows only HTTP packets)
```

```
tcp.port == 80      (shows only TCP port 80 traffic)
```

```
ip.addr == 192.168.1.100      (shows traffic to/from specific IP)
```

Clear filters by clicking 'Expression' button and selecting 'Reset'.

Key Observations

- Protocol Column: Shows the protocol used (HTTP, DNS, TCP, etc.)
- Source/Destination: Shows IP addresses and ports
- Info Column: Shows details about the packet content
- By expanding protocol layers in the middle pane, you can see individual field values

Activity 2: Identifying Protocol Layers

Objective: Understand the encapsulation of data across network layers

- Capture an HTTP packet (follow previous steps)
- Click on an HTTP packet to select it
- In the middle pane, you will see protocol layers (expand each by clicking the + sign):
- Frame (Physical/Data Link), Ethernet (Data Link), IP (Network), TCP (Transport), HTTP (Application)

What to Observe

- MAC addresses in Ethernet header
- IP addresses and protocol numbers in IP header
- Source and destination ports in TCP header
- HTTP request/response data in the application layer

PART 3: ARP SPOOFING WITH ETTERCAP

What is Ettercap?

Ettercap is a comprehensive suite for Man-in-the-Middle (MITM) attacks. It can perform ARP spoofing, DNS spoofing, and traffic filtering/modification. Ettercap provides both command-line and graphical interfaces.

Activity 3: ARP Spoofing Attack (Lab Environment Only)

CRITICAL WARNING: This attack should ONLY be performed in your isolated lab network. Unauthorized use on real networks is illegal.

Setup: You need at least two machines on your lab network - one attacker machine and one victim machine.

Step 1: Identify Your Lab Network

On your attacker machine, find the network configuration:

```
ifconfig      (on Linux/Mac) or ipconfig (on Windows)
```

Note your IP address and the network range. For example:

```
inet 192.168.122.100 netmask 255.255.255.0
```

Step 2: Scan the Network for Targets

Discover what devices are on your lab network. Use nmap to scan:

```
sudo nmap -sn 192.168.122.0/24
```

This will show all active devices on the network. Note the victim machine's IP address (e.g., 192.168.122.50).

Step 3: Launch Ettercap

```
sudo ettercap -G      (for graphical interface)
```

Or in command-line:

```
sudo ettercap -T -i eth0
```

Step 4: Configure Ettercap for ARP Spoofing (GUI Version)

- Select interface: Sniffing → Unified Sniffing → Select your network interface
- Scan hosts: Hosts → Scan for hosts (wait for scan to complete)
- View hosts: Hosts → Host list (see all discovered devices)
- Select targets: Click on victim machine IP, then 'Add to Target 1'
- Also add the gateway IP (usually .1 address) to 'Add to Target 2'

Step 5: Start MITM Attack

- Click: Mitm → Arp Spoofing
- A dialog will appear. Leave options as default and click OK
- Ettercap now begins spoofing ARP packets

Step 6: Start Sniffing

- Click: Start → Start Sniffing
- Any traffic between victim and gateway will now pass through your machine
- Captured packets will appear in Ettercap's display pane

Step 7: Monitor Captured Traffic

- On your victim machine, try accessing websites or running network applications
- Watch Ettercap capture this traffic
- Note: HTTPS traffic will be encrypted, but metadata (domains, IP addresses) will still be visible

Step 8: Stop the Attack

- Click: Start → Stop Sniffing
- Click: Mitm → Arp Spoofing (again to stop)
- This restores normal network operation for the victim

PART 4: ADVANCED ACTIVITIES

Activity 4: DNS Spoofing with Ettercap

DNS spoofing redirects victims to fake websites. In Ettercap:

- Click: Plugins → Manage Plugins
- Find 'dns_spoof' and activate it
- Edit /etc/ettercap/etter.dns file to specify domain redirections:

```
example.com 192.168.122.100    (redirects example.com to your machine)
```

- Start ARP spoofing and sniffing (as in Activity 3)
- When victim tries to access example.com, they will be redirected to your machine

Activity 5: Capturing Credentials with Wireshark

After performing MITM attack with Ettercap:

- Start Wireshark packet capture on your attacker machine
- Have victim access an HTTP-based email service or website login page
- Filter for HTTP traffic: http in Wireshark
- Look for POST requests, which contain form data
- Right-click packet → Follow → HTTP Stream
- You can see the submitted credentials in plaintext (if HTTP, not HTTPS)

Activity 6: SSL Stripping Attack (Optional - Advanced)

SSL stripping forces HTTPS connections to downgrade to HTTP. Using mitmproxy:

```
sudo mitmproxy -T --mode transparent
```

This is an advanced attack that captures HTTPS traffic when combined with ARP spoofing. Consult advanced resources for detailed setup.

Activity 7: Detecting Sniffing Attacks

Learn to identify when your network is under attack:

- Monitor ARP traffic with Wireshark: arp in filter
- Look for duplicate ARP replies from different MAC addresses
- Use tools: arpswatch or arp-scan to monitor changes in ARP tables
- Unusual high volume of ARP packets indicates potential ARP spoofing

PART 5: ASSIGNMENT - PACKET ANALYSIS PROJECT

Project Overview

For this assignment, you will perform a complete packet analysis capture and document your findings. This project demonstrates your understanding of network protocols and the ability to analyze traffic for security purposes.

Assignment Requirements

Part 1: Packet Capture

1. Capture at least 100 packets using Wireshark on your lab network
2. Include traffic from multiple protocols (HTTP, DNS, TCP, ARP, etc.)
3. Save the capture file as pcap format
4. Duration: 2-5 minutes of continuous capture

Part 2: Analysis Report

5. Create a document (Word or PDF) with the following sections:

Section 1: Capture Overview

- Date and time of capture
- Network interface used
- Total number of packets captured
- Brief description of network activity during capture

Section 2: Protocol Distribution

- Identify the different protocols in your capture
- Count how many packets for each protocol (create a table)
- Example: HTTP (25 packets), DNS (15 packets), TCP (30 packets), etc.
- Use Wireshark's Statistics menu to help count

Section 3: Detailed Packet Analysis

- Select 3-5 interesting packets from your capture
- For each packet, document:
 - Source and destination IP addresses
 - Source and destination MAC addresses
 - Protocol used
 - Port numbers (if applicable)
 - Key information from the packet
 - Which OSI layer(s) this packet operates on

Section 4: Security Observations

- Identify any unencrypted protocols in your capture (HTTP, Telnet, FTP, etc.)
- Explain why unencrypted protocols are vulnerable to sniffing
- Identify HTTPS/TLS traffic (encrypted traffic you CANNOT read)

- Describe what information is still visible even in encrypted traffic
- List potential vulnerabilities exposed in your traffic

Section 5: Defense Recommendations

- Based on your analysis, what protections should be implemented?
- Describe at least 3 mitigation strategies
- Explain how each mitigation would prevent the attacks you studied
- Include at least one technical recommendation for network configuration

Submission Requirements

6. Save analysis report as Word (.docx) or PDF
7. Include screenshots of Wireshark showing your capture and analysis
8. Submit PCAP file separately
9. Report should be 3-5 pages with proper formatting
10. Include a header with: Your name, Date, Course, and 'Lab Assignment: Packet Analysis'
11. Ensure all information is accurate and explanations are clear

TROUBLESHOOTING GUIDE

Wireshark Issues

Problem: Cannot capture packets

Solution: Run with sudo, select correct interface, ensure capture interface has activity

Problem: Wireshark shows 'Unknown' protocols

Solution: This is normal. Expand the packet layers to see actual protocols. Some traffic may use non-standard ports.

Ettercap Issues

Problem: ARP spoofing not working

Solution: Ensure both target IPs are correctly set, that you're using root access, check interface selection

Problem: Victim still connected to network

Solution: This is correct behavior. MITM position doesn't disconnect victim. You intercept their traffic.

WIRESHARK FILTER REFERENCE

Common filters to use in Wireshark for efficient packet analysis:

Filter	Purpose
<code>http</code>	Show only HTTP traffic
<code>dns</code>	Show only DNS traffic
<code>arp</code>	Show only ARP packets
<code>tcp.port==80</code>	Show only traffic on port 80 (HTTP)

Filter	Purpose
<code>tcp.port==443</code>	Show only traffic on port 443 (HTTPS)
<code>ip.addr==192.168.1.1</code>	Show traffic to/from specific IP
<code>tcp</code>	Show only TCP packets
<code>udp</code>	Show only UDP packets

ADDITIONAL RESOURCES AND REFERENCES

- Wireshark Official Documentation: <https://wireshark.org/docs/>
- Ettercap Project: <https://www.ettercap-project.org/>
- mitmproxy Documentation: <https://mitmproxy.org/>
- OWASP Testing Guide: <https://owasp.org/www-project-web-security-testing-guide/>
- RFC 791 - Internet Protocol (IP): <https://tools.ietf.org/html/rfc791>
- RFC 793 - Transmission Control Protocol (TCP): <https://tools.ietf.org/html/rfc793>
- RFC 826 - Address Resolution Protocol (ARP): <https://tools.ietf.org/html/rfc826>